SARTHAK SHRIVASTAVA

Vindhyachal House, IIT Delhi, Hauz Khas, New Delhi, India - 110016

sarthaks20.github.io | sarthakshrivastava | sarthak.iitd23@gmail.com | (+91)-903-966-0358

EDUCATION

Indian Institute of Technology Delhi, India *2019 - 2023*
Bachelor of Technology (B.Tech.), Civil Engineering *GPA: 8.283/10.0*

AREAS OF INTEREST

Computational Structural Mechanics, Continuum Mechanics, Finite Element Methods, Multi-scale and Multi-physics Modeling, Fracture Mechanics, Peridynamics, Molecular Dynamics, Structural Dynamics, Structural Health Monitoring, Numerical Methods, Machine Learning in Mechanics

RESEARCH EXPERIENCE

Research Assistant | *Multiphysics & Multiscale Mechanics Research Group (M3RG), IIT Delhi* (*Jan, 2023 - Present*)

- Conducting numerical simulations using finite element and peridynamics methods to investigate the role of indenter geometry on indentation-induced damage of glasses under the supervision of Prof. N. M. Anoop Krishnan

Summer Undergraduate Research Fellow | *Industrial Research & Development Unit, IIT Delhi* (*May, 2021 - Oct, 2021*)

- Worked on a summer research project titled 'The Application of Machine Learning to Structural Health Monitoring' under the supervision of Prof. Sahil Bansal, funded by the Industrial Research & Development (IRD) Unit, IIT Delhi

PROJECTS

Peridynamics-Based Modeling of Nanoindentation Response of Silicate Glasses (*Sep, 2022 - Present*)
Advisor: Prof. N. M. Anoop Krishnan, IIT Delhi

- Developing a **peridynamics**-based numerical model to investigate the nanoindentation behavior of silicate glass under **varying tip angles of the indenter**, providing insights into the nanoscale mechanical behavior of the material
- Validating the numerical model using **finite element analysis** (FEA), demonstrating the accuracy and reliability of the proposed model in predicting the **nanoindentation response** of silicate glasses
- Conducting **upscaling analysis** to investigate the macroscopic behavior of silicate glass under different indentation conditions, providing a better understanding of the mechanical properties of the material at the macroscale level

Stability of Steel Columns at Elevated Temperatures | *Finite Element Analysis* (*Nov, 2022 - Dec, 2022*)
Advisor: Prof. Vasant Matsagar, IIT Delhi

- Eigen value problem to determine the **buckling load** through FE formulation showed convergence with Euler's formula
- Stresses were evaluated under **thermo-mechanical loading** for various boundary conditions of the column
- Performed **non-linear analysis** of columns using **Riks analysis**; determined **critical thermo-mechanical stresses** and facilitated the performance of **post-buckling analysis** until collapse

The Application of Machine Learning to Structural Health Monitoring | *UROP* (*May, 2021 - Oct, 2021*)
Advisor: Prof. Sahil Bansal, IIT Delhi

- Devised a **vibration-based damage-detection** approach using **AI methods** on time response simulated data
- Extracted different **damage-sensitive features** based on **signal statistics** and **modal properties**; analyzed **signal response time-series** using **AR and ARX models**; proposed **statistical model for feature discrimination**
- Ranked damage-sensitive features and employed **support vector machines** to classify **structural condition states**

Transverse Vibration Analysis of Euler-Bernoulli Beam | *Computational Mechanics* (*Mar, 2022 - Apr, 2022*)
Advisor: Prof. Prapanch Nair, IIT Delhi

- Derived an **explicit finite difference scheme** with **second order accuracy** by discretizing the governing equations
- Implemented Python script for computation of **natural frequencies** of vibration, **mode shapes**, and **vibration response** of the elastic beam for **different boundary conditions**, viz. fixed-fixed, fixed-pinned, pinned-pinned, and fixed-free
- **von Neumann analysis** was done to establish **stability requirements** of the solution, explicit FD scheme was found to be **conditionally stable** and **convergent**; compared the results with analytical solution to validate the approach

Structural Design using STAAD.Pro | *Structural Design & Detailing* (*Sep, 2022 - Nov, 2022*)

- Designed and analyzed the structures of **hostel campus** comprising five blocks of **G+10 storey buildings**, **reinforced concrete overhead water tank**, **industrial shed truss**, and **transmission tower** using AutoCAD and STAAD.Pro
- Resistance to **earthquake** and **wind loads** were incorporated into the structural design process as per Indian Standards

Food Outlet Management System | *Data Structures & Algorithms* (*Apr, 2021*)

- Developed a simulation environment for a food outlet having facility of **multiple ordering/billing counters** and food delivery on **first-order-first-serve basis**; implemented **Priority Queue**, **AVL Tree**, **Array Deque** based data structures
- Statistics like **average waiting time**, **average queue length** were computed in order to **improve customer service**

TEACHING & MENTORING EXPERIENCE

Tutor | *Course Hero, Inc.* (*Dec, 2021 - Apr, 2023*)
- Offered online tutoring to students seeking academic assistance in Physics, Mathematics, and Civil Engineering
- Received **consistently positive ratings** and feedback from students for effectiveness and clarity of instruction

Academic Mentor | *Board of Student Welfare (BSW), IIT Delhi* (*Nov, 2020 - Feb, 2021*)
- Provided **peer mentorship** and tutoring to undergraduate freshers in **Engineering Mechanics** (APL100) course
- Assisted the students to acclimate to a new academic setting by **facilitating weekly study sessions**

Educator (IIT-JEE Physics) | *Unacademy* (*Aug, 2020 - Oct, 2020*)
- Taught Advanced Physics to **500+ IIT-JEE aspirants**; held interactive sessions and emphasized **analytical thinking**, **problem-solving** and **exam-taking** skills through the use of **visual learning** approaches; offered learning resources
- Ranked among **top 100 educators** in IIT-JEE category with over **15K minutes** of watch time on online platforms

Mentor | *Project Aarohan, National Service Scheme (NSS), IIT Delhi* (*Sep, 2019 - Apr, 2020*)
- Mentored underprivileged students of class XI and XII for competitive exams
- Taught Physics; provided them with comprehensive study material and conducted doubt clearing sessions

WORK EXPERIENCE

Summer Analyst | *Axis Bank (Business Intelligence Unit)* (*Jun, 2022 - Jul, 2022*)
- Built a **web-scraping tool** that extracts information related to employees' salaries, revenues, capitals of an organization
- Employed **Selenium** to **automate web browser** and interact with **dynamic web pages**; handled **captcha recognition** through **OpenCV** based **character segmentation** technique; analyzed the scraped data to gain useful **business insights**
- Developed tool is capable of scraping data of **600K+ organizations**; proposed ideas to **amplify sales** of salary accounts

TECHNICAL SKILLS

- **Programming Languages:** C++, Python, Java, Julia, MATLAB
- **Software & Tools:** Microsoft Office, High Performance Computing, STAAD.Pro, Abaqus FEA, Ansys, LaTeX, Git
- **Libraries:** C++ STL, Pandas, NumPy, SciPy, Matplotlib, Scikit-learn, Peridigm, Selenium, Tesseract-OCR, OpenCV

SCHOLASTIC ACHIEVEMENTS

- Awarded the **Summer Undergraduate Research Award (SURA), 2021** from Industrial R&D Unit, IIT Delhi *2022*
- **Online Judges:** Scored **5-star rating** on ***HackerRank*** and **3-star (max. 1783 rating)** on ***CodeChef*** *2021*
- **JEE (Advanced):** Secured **All India Rank 1841 (top 1 percentile)** among 0.17 million candidates *2019*
- **JEE (Main):** Achieved **All India Rank 1403 (top 0.1 percentile)** among 1.2 million candidates *2019*

PRESENTATIONS

Impact Studies with Peridynamics, Civil Engineering Seminar, Indian Institute of Technology, Delhi, India. Apr 2022.

REFERENCES

Dr. N. M. Anoop Krishnan
Associate Professor
Department of Civil Engineering
Indian Institute of Technology Delhi
krishnan@civil.iitd.ac.in
https://sites.google.com/site/nmakrishnaniitd/
+91-11-2659-1223

Dr. Vasant Matsagar
Professor
Department of Civil Engineering
Indian Institute of Technology Delhi
matsagar@civil.iitd.ac.in
http://www.matsagar.org/
+91-11-2659-1225

Dr. Sahil Bansal
Assistant Professor
Department of Civil Engineering
Indian Institute of Technology Delhi
sahil@civil.iitd.ac.in
https://sites.google.com/view/sahilbansal/
+91-11-2654-8414

Dr. Prapanch Nair
Assistant Professor
Department of Applied Mechanics
Indian Institute of Technology Delhi
pnair@am.iitd.ac.in
https://web.iitd.ac.in/~pnair
+91-11-2659-1226